

TOC-2025: Quiz-1 (DFA Concepts)

Full Marks: 15

(1) Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA and δ be its transition function. The *extended transition function* $\hat{\delta} : Q \times \Sigma^* \rightarrow Q$ is defined as

- $\hat{\delta}(q, \varepsilon) = q$ and
- If w is a string of the form xa such that $a \in \Sigma$ and $x \in \Sigma^*$, then $\hat{\delta}(q, xa) = \delta(\hat{\delta}(q, x), a)$.

Use this definition to prove that $\hat{\delta}(\hat{\delta}(q, x), y) = \hat{\delta}(q, xy)$.

(8 Marks)

SOLUTION. We prove the statement by induction on the length of $y \in \Sigma^*$.

Basis Step: If $|y| = 0$, then $y = \varepsilon$. Hence,

$$\begin{aligned}\hat{\delta}(\hat{\delta}(q, x), y) &= \hat{\delta}(\hat{\delta}(q, x), \varepsilon) \\ &= \hat{\delta}(q, x) \quad (\text{by definition of } \hat{\delta}) \\ &= \hat{\delta}(q, x\varepsilon) \\ &= \hat{\delta}(q, xy).\end{aligned}$$

Induction Hypothesis: Assume that for some $k \geq 0$,

$$\hat{\delta}(\hat{\delta}(q, x), y) = \hat{\delta}(q, xy)$$

holds $\forall x \in \Sigma^*$ and $\forall y \in \Sigma^*$ such that $|y| = k$.

Induction Step: Let $|y| = k + 1$. Then $y = wa$, where $w \in \Sigma^*$ with $|w| = k$ and $a \in \Sigma$. Thus,

$$\begin{aligned}\hat{\delta}(\hat{\delta}(q, x), y) &= \hat{\delta}(\hat{\delta}(q, x), wa) \\ &= \delta(\hat{\delta}(\hat{\delta}(q, x), w), a) \quad (\text{by definition of } \hat{\delta}) \\ &= \delta(\hat{\delta}(q, xw), a) \quad (\text{by induction hypothesis}) \\ &= \hat{\delta}(q, xwa) \quad (\text{by definition of } \hat{\delta}) \\ &= \hat{\delta}(q, xy).\end{aligned}$$

Therefore, by the principle of mathematical induction,

$$\hat{\delta}(\hat{\delta}(q, x), y) = \hat{\delta}(q, xy)$$

holds for all $x, y \in \Sigma^*$.

(2) Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA and $p_1, p_2 \in Q$ be two distinct states such that for all input symbols $a \in \Sigma$,

- $\delta(p_1, a) = p_2$, and
- $\delta(p_2, a) = p_2$.

Prove (by induction) or disprove that for all strings $w \in \Sigma^*$, $\widehat{\delta}(p_1, w) = p_2$. (7 Marks)

SOLUTION. We DISPROVE the statement.

It suffices to show that there exists some $w \in \Sigma^*$ for which the claim does not hold.

Since $\varepsilon \in \Sigma^*$, consider $w = \varepsilon$. By the definition of the extended transition function,

$$\widehat{\delta}(p_1, w) = \widehat{\delta}(p_1, \varepsilon) = p_1.$$

Given that $p_1 \neq p_2$, we have

$$\widehat{\delta}(p_1, \varepsilon) \neq p_2.$$

Hence, the statement is false.

Please note that the choice $w = \varepsilon$ is the only counterexample. **No marks** are awarded for proposing a non-empty string $w \in \Sigma^*$, since the statement holds true for all such strings.